SUPERSEDING
MIL-M-81385(AS)
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MILITARY SPECIFICATION

# MANIFOLD STEM ASSEMBLIES, LIFE PRESERVER

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope - This specification covers the requirements for one type of rubber sleeve check valve and four types of life preserver manifold stern assemblies with or without an installed rubber sleeve check valve.

1.2 Classification - The manifold stem assemblies shall be of the following types, as specified (see 6. 2):

Type I (62A120H1-1)
Type II (62A120H1-2)
Type III (52A6598)
Type IV (61B4286)

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-566 | Boxes, Folding, Paperboard |
| PPP-B-636 | Box, Fiberboard |
| PPP-B-676 | Boxes, Set-up |
| PPP-C-843 | Cushioning Material, Cellulosic |
| PPP-T-45 | Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing |



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Military

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of |
| MIL-C-601 | Carbon Dioxide, Technical, Cylinders of, for Inflation of Life Preserver |
| MIL-A-5540 | Adhesive, Polychloroprene |
| MIL-I-23145 | Inflation Assemblies, Life Preserver |
| MIL-C-25369 | Cylinder, Carbon Dioxide, for Inflation of Pneumatic Life Preservers |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-151 | Metals, Test Methods |
| FED-STD-601 | Rubber, Sampling and Testing |

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |

DRAWINGS

Bureau of Naval Weapons

| | |
|---|---|
| 62A120H1 | Manifold Stem Assembly and Details, for Life Preservers |

Bureau of Ships

| | |
|---|---|
| S3306-1183395 | Rubber Sleeve Check Valve for $C0_2$ Inflatable Life Preserver |

Air Force

| | |
|---|---|
| 52A6598 | Stem-Inflator Valve, Life Preserver Bladder |
| 52C6599 | Insert-Inflator Valve Stem, Life Preserver Bladder |

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Air Force (Continued)

| | |
|---|---|
| 52A6600 | Cap-Inflator Valve, Pneumatic Life Preserver |
| 61B4286 | Stem-Inflator Valve, Life Preserver Cell, Assembly of |
| 61C4287 | Insert-Inflator Valve Stem, Life Preserver Cell |

(When requesting any of the applicable documents, refer to both title and number. Copies of the applicable documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. )

2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H28 ScrewThread Standards for Federal Services

(Copies of Handbook H28 maybe obtained from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. )

3, REQUIREMENTS

3.1 First article - Unless otherwise specified, the rubber sleeve check valves or the manifold stem assemblies, with or without an installed rubber sleeve check valve, as applicable, furnished under this specification shall be a product which has been inspected and has passed the first article inspection specified in Section 4.

3.2 Materials and components - The materials and components shall conform to the applicable specifications and drawings as listed or required herein. Unless otherwise specified, the elastomeric components used in the construction of the rubber sleeve check valves and manifold stern assemblies shall have been manufactured not more than 12 months prior to the date of delivery of the rubber sleeve check valves or the manifold stem assemblies, with or without an installed rubber sleeve check valve, as applicable (see 6. 2).

3.3 Design and construction -



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3. 3,1 Rubber sleeve check valve- The rubber sleeve check valve shall conform to S3306-1183395. The rubber sleeve shall be fabricated without any plasticizer or mold lubricant. When installed on the check valve, the rubber sleeve or check valve shall not contain any lubricant.

3. 3.2 Manifold stem assemblies - The manifold stem assemblies shall conform to 62A120H1-1 or 62A120H1-2, 52A6598, and 61B4286, as applicable. The rubber base shall be securely molded to the stem, The bonding material shall not affect the rubber base or metal and shall not exude or become stiff or brittle. The composition of the base shall be such that it shall have good cementability properties, when used with the adhesive conforming to MIL-A-5540, Class 1, 2, or 3. The rubber base, if desired, may be lightly buffed, on both sides, prior to being cemented to the life preserver. When specified, the manifold stem assemblies shall be furnished with the rubber sleeve check valve installed on the inside of the manifold stem (see 6. 2),

3. 3.3 Threads - The threads on the components of the manifold stem assemblies and rubber sleeve check valve shall conform to 62A120H1, S3306-1183395, 52C6599, 52A6600, and 61C4287, as applicable, and H28,

3.4 Performance -

3. 4.1 Strength of the attachment of the molded rubber base to the manifold stem -

3.4.1.1 As received - When inspected as specified in 4.6.3.1, the attachment of the molded rubber base to the brass manifoid stem shall withstand a pull of 250 pounds without separating.

3.4. 1.2 After aging - When inspected as specified in 4.6.3.2, the attachment of the molded rubber base to the brass manifold stem shall withstand a pull of 185 pounds without separating.

3. 4.2 Back pressure leakage - When inspected as specified in 4.6.4, the manifold stem assembly shall not leak.

3. 4.3 Operation - When inspected as specified in 4.6.5, the hole in the side of the stem shall align with one of the holes in the outlet of the inflation assembly and the gas shall not pass through any part of the manifold stem assembly into the atmosphere other than at the base or outlet side of the stem and when applicable, the rubber sleeve of the check valve shall not be dislodged. When the manifold stem assembly is purchased with an installed check valve, the manifold stem assembly shall then be inspected and conform to the requirements for the back pressure leakage, 3.4.2.

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3.4.4 Salt spray - When exposed to salt spray as specified in 4.6.6, the rubber sleeve check valve or the manifold stem assembly, with or without an installed rubber sleeve check valve, as applicable, shall show no appreciable corrosion or other damage. The cap nut, Type I base nut, and the rubber sleeve check valve, as applicable, shall not become **frozen** to the stem and shall be capable of being removed without difficulty,

3.5 Marking - The rubber base, after buffing when applicable, shall be marked on both sides with the date of cure (month and year). Unless otherwise specified, the markings shall be legible and durable yellow or white numerals which shall be thoroughly dry prior to packaging. The marking medium shall not contain any material which is deleterious to the rubber base nor shall the rubber base be damaged by the method of marking.

3.6 Workmanship - The check valves or the manifold stem assemblies, with or without an installed check valve, as applicable, shall not be deformed or misaligned nor contain any rough surface, nick, burr, sharp edge, crack, embedded foreign matter, loose or detached component, or lubricant. The rubber components shall not contain any hole, tear, cut, patch, mend, burn, or abraded area. Because of the emergency and life support use of this equipment, the importance of providing a product of uniform excellent quality cannot be overemphasized. The manifold stem assemblies or check valves, as applicable, shall be uniform in quality and shall be free from irregularities or defects which could adversely affect performance, reliability, or durability. The check valves or the manifold stem assemblies, with or without an installed check valve, as applicable, shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The examination and testing of the rubber sleeve check valves or the manifold stem assemblies, with or without an installed check valve, as applicable, shall be classified as follows:

(a) First article inspection - First article inspection consists of examinations and tests performed on samples which are representative of the production item after the award of a contract to determine that the production item meets the re-



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quirements of this specification.

(B) Quality conformance inspection - Quality conformance inspection consists of examinations and tests performed on individual products *or* lots to determine conformance of the products or lots with the requirements set forth in this specification.

4.3 First article inspection - The first article inspection of the rubber sleeve check valves or the manifold stem assemblies, with or without an installed check valve, as applicable, shall consist of examinations and tests for all of the requirements of this specification.

4. 3,1 First article samples - Unless otherwise specified, as soon as practicable after the award of the contract or order, the manufacturer shall submit six rubber sleeve check valves or six manifold stem assemblies of each type specified in the contract or order, with or without an installed check valve, as applicable. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a contractor is in continuous production of these rubber sleeve check valves or the manifold stem assemblies, with or without an installed check valve, as applicable, from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the procuring activity. Approval of the first article inspection samples or the waiving of the first article inspection does not preclude the requirements for performing the quality conformance inspection. The first article inspection samples shall be forwarded as specified in the contract or order.

4.3.2 Upon completion of the first article inspection, all the applicable inspection reports and when applicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the Government quality assurance representative. One approved rubber sleeve check valve or manifold stem assembly, with or without an installed check valve, as applicable, will be returned to the manufacturer for use in monitoring production. The other five check valves or five manifold stem assemblies, with or without an installed check valve, as applicable, will be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract.

4.4 Quality conformance inspection - The sampling and inspection levels shall conform to MIL-STD-105. The quality conformance inspection shall consist of the following:

Amount of natural rubber in the check valve rubber sleeve
Visual examination
Dimensional check
Attachment of the molded robber base to the manifold stem
as received and after aging



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Back pressure leakage
Operation
Preparation for delivery

4. 4.1 Sampling -

4. 4.1.1 Inspection lot-

4. 4.1.1.1 Check valves - An inspection lot size shall be expressed in units of one rubber sleeve check valve made essentially under the same conditions and from the same materials and components. The sample unit shall be one rubber sleeve check valve.

4. 4.1.1.2 Manifold stem assemblies - An inspection lot size shall be expressed in units of one manifold stem assembly of one type, with or without an installed check valve, as applicable, made essentially under the same conditions and from the same materials and components. The sample unit shall be one manifold stem assembly of one type, with or without an installed check valve, ***as*** applicable.

4. 4.1.1.3 Preparation for delivery - An inspection lot size shall be expressed in units of one fully prepared shipping container, containing rubber sleeve check valves or manifold stem assemblies of one type, with or without an installed check valve, as applicable, fully prepared for delivery from essentially the same materials and components. The sample unit shall be one shipping container, containing rubber sleeve check valves or manifold stem assemblies of one type, with or without an installed check valve, as applicable, fully prepared for delivery with the exception that it need not be sealed.

4. 4.1.2 Sampling for tests and examinations of the rubber sleeve check valve and the manifold stem assemblies - The sample size, acceptance criteria, tests, and examinations required for the rubber sleeve check valves and the manifold stem assemblies, with or without an installed check valve, as applicable, shall be as specified in Table I.

4.5 Inspection conditions -

4. 5.1 Atmospheric conditions - Unless otherwise specified, all inspections required by this specification shall be conducted at an atmospheric pressure of 28 to 32 inches of mercury, at a temperature of 77 $\pm$ 18 degrees Fahrenheit (25 $\pm$ 10 degrees Centigrade), and a relative humidity of 80 percent or less. When the inspections are conducted at an atmospheric pressure or temperature different from the above values, proper corrections shall be made for the change in instrument reading.



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TABLE I

SAMPLE SIZE, ACCEPTANCE CRITERIA, TESTS, AND EXAMINATIONS OF THE RUBBER SLEEVE CHECK VALVES AND THE MANIFOLD STEM ASSEMBLES

| INSPECTION | PARAGRAPH | | SAMPLE SIZE | ACCEPTANCE CRITERIA |
|---|---|---|---|---|
| | REQUIREMENT | METHOD | | |
| Natural rubber content in the check valve rubber sleeve 1/ | 3.3.1 | 4.6.1 | Inspection Level S-1 2/ | Acceptance number zero, rejection number 1 |
| Visual examination | Section 3, 62A120H1, S3306-1183395, 52A6598, 52C6599, 52A6600, 61B4286, and 61C4287, as applicable | 4.6.2.1 | Every rubber sleeve check valve and manifold stem assembly, as applicable | Reject all units with any defect |
| Dimensional check | Section 3, 62A120H1, S3306-1183395, 52A6598, 52C6599, 52A6600, 61B4286, and 61C4287, as applicable | 4.6.2.1 | Inspection Level S-3 | An acceptable quality level of 4.0 defects per 100 units |
| Strength of the attachment of the molded rubber base to the manifold stem as received | 3.4.1.1 | 4.6.3.1 | Inspection Level S-1 2/ | Acceptance number zero, rejection number 1 |
| Strength of the attachment of the molded rubber base to the manifold stem after aging | 3.4.1.2 | 4.6.3.2 | Inspection Level S-1 2/ | Acceptance number zero, rejection number 1 |
| Back pressure leakage 3/ | 3.4.2 | 4.6.4 | Inspection Level S-1 2/ | Acceptance number zero, rejection number 1 |
| Operation | 3.4.3 | 4.6.5 | Inspection Level S-1 2/ | Acceptance number zero, rejection number 1 |
| Preparation for delivery | Section 5 | 4.6.2.2 | Inspection Level S-2 | An acceptable quality level of 2.5 defects per 100 units |

1/ This inspection is applicable when the rubber sleeve check valves are purchased individually and when the manifold stem assemblies are purchased with an installed check valve.

2/ The sample size shall be based only on the applicable sample size code letter corresponding to the specified inspection level of MIL-STD-105.

3/ This inspection is applicable only when the manifold stem assemblies are purchased with an instalied rubber sleeve check valve.



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4. 5.2 inspection area and equipment- The area in which the check valves or the manifold stem assemblies, with or without an installed check valve, as applicable, are inspected shall be adequately protected to preclude damage to the units. The area and inspection equipment shall be free of sharp or rough edges, burrs, protrusions, etc., which will cut, tear, or damage the chock valves or the manifold stem assembly, with or without m" installed check valve, as applicable,

4.6 Inspection methods -

4.6.1 Natural rubber content in the check valve rubber sleeve- The natural rubber content in the check valve rubber sleeve shall be determined in accordance with FED-STD-601, Method 16121.

4.6.2 Visual examination -

4.6.2.1 Check valves and manifold stem assemblies - Every check valve or every manifold stem assembly, with or without an installed check valve, as applicable, shall be examined visually to determine conformance to this specification. Each check valve or each manifold stem assembly, with or without an installed check valve, as applicable, selected as a sample unit from the lot, shall be thoroughly checked dimensionally to determine conformance to this specification. The lists of defects, Tables II and III, shall be used to enumerate the defects found.

TABLE 11

LIST OF DEFECTS FOR VISUAL EXAMINATION
OF THE CHECK VALVES AND MANIFOLD STEM ASSEMBLIES

| DEFECT |
|---|
| 1. Any surface rough, misaligned, or contains any nick, sharp edge, crack, burr, embedded foreign matter, dent, or pit mark. |
| 2. Any finish missing or any component improperly finished, as applicable. |
| 3.. Any component missing, malformed, corroded, fractured, bent, distorted, or damaged. |
| 4. Any component loose, detached, not attached as specified, or otherwise not securely retained. |
| 5. Any threads missing, stripped, torn, broken, or otherwise defective. |
| 6. Any hole, tear, cut, patch, burn, or abraded area on any rubber component . |



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TABLE II (Continued)

| DEFECT |
|---|
| 7.. Any plasticizer or lubricant on the rubber sleeve of the check valve or on the check valve 1/. |
| 8. Any lubricant on the manifold stem. |
| 9. Check valve does not fit in the manifold stem due to defect of the valve or defect of the stem, not inserted properly in the manifold stem, or the manifold stem does not contain an installed check valve 2/. |
| 10. Not specified type of manifold stem assembly. |
| 11. Design of any component not as specified. |
| 12. Cure date missing, incorrect, illegible, incomplete, or improperly located on the rubber base. |

1/ Applicable when the check valve is purchased individually and when the manifold stem assemblies are purchased with an installed check valve.

2/ Applicable only when the manifold stem assemblies are purchased with an installed check valve.

TABLE III

LIST OF DEFECTS FOR THE FINISHED DIMENSIONS
OF THE CHECK VALVES AND MANIFOLD STEM ASSEMBLIES

| EXAMINE | DEFECT |
|---|---|
| Measure the check valve and all the components of the manifold stem assembly, as applicable | Any measurement deviating from the dimensions and tolerances as specified in 62A120H1, S3306-1183395, 52A6598, 52 C6599, 52 A6600, 61B4286, and 61C4287, as applicable, shall be enumerated as a dimensional defect. |

4.6.2.2 Preparation for delivery - Each of the fully prepared shipping containers, containing rubber sleeve check valves or manifold stem assemblies of one type, with or without an installed check valve, as applicable, selected as a sample unit from the lot, shall be visually examined to determine that the packaging, packing, and marking conform to this specification. The list of defects, Table IV, shall be used to enumerate the defects found.



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TABLE IV

LIST OF DEFECTS FOR PREPARATION FOR DELIVERY

| ITEM | DEFECTS |
|---|---|
| Exterior and interior markings | Missing, incorrect, incomplete, illegible; of improper size, location, sequence, or method of application; markings not the same on the interior and exterior containers; information on the paper label in the film bag not readable through the film or no label in the film bag. |
| Packaging and packing material | Any non-conforming component; any component missing, damaged, or otherwise defective. |
| Workmanship | Inadequate application of the components such as incomplete closure of the film bag, unit package, container flaps, loose strappings, etc. ; bulging or distortion of the containers; any portion of the manifold stem assembly or check valve, as applicable, distorted or caught in the heat sealed seam. |
| Exterior and interior weight or content | Number per container is more or less than required; gross or net weight exceeds the requirements; more than one type in the same container; check valve, when a component of the manifold stem assembly, is packaged or packed separately. |

4.6.3 Strength of the attachment of the molded rubber base to the manifold stem - Each manifold stem assembly shall be inspected without the cap nut, gasket, washer, Type I base nut, or check valve, as applicable, installed. Upon completion of the attachment of the molded rubber base to the manifold stem as received and after aging inspections, the rubber bases and stems shall be discarded and shall not be part of the quantity to be delivered under contract,

4.6. 3.1 As received - The rubber base shall be centered and secured in an adapter having 1-1/4 inch inside diameter opening around the stem as in Figure 1. The adapter shall be attached to the lower jaw of the inspection apparatus. An attachment shall be screwed into or attached to the top of the manifold stem and then attached to the upper jaw of the inspection apparatus. The direction of the pull shall be perpendicular to the molded rubber base. The speed of the movable jaw of the inspection apparatus under no load shall be 12 $\pm$ 1/2 inches per minute. The jaws shall



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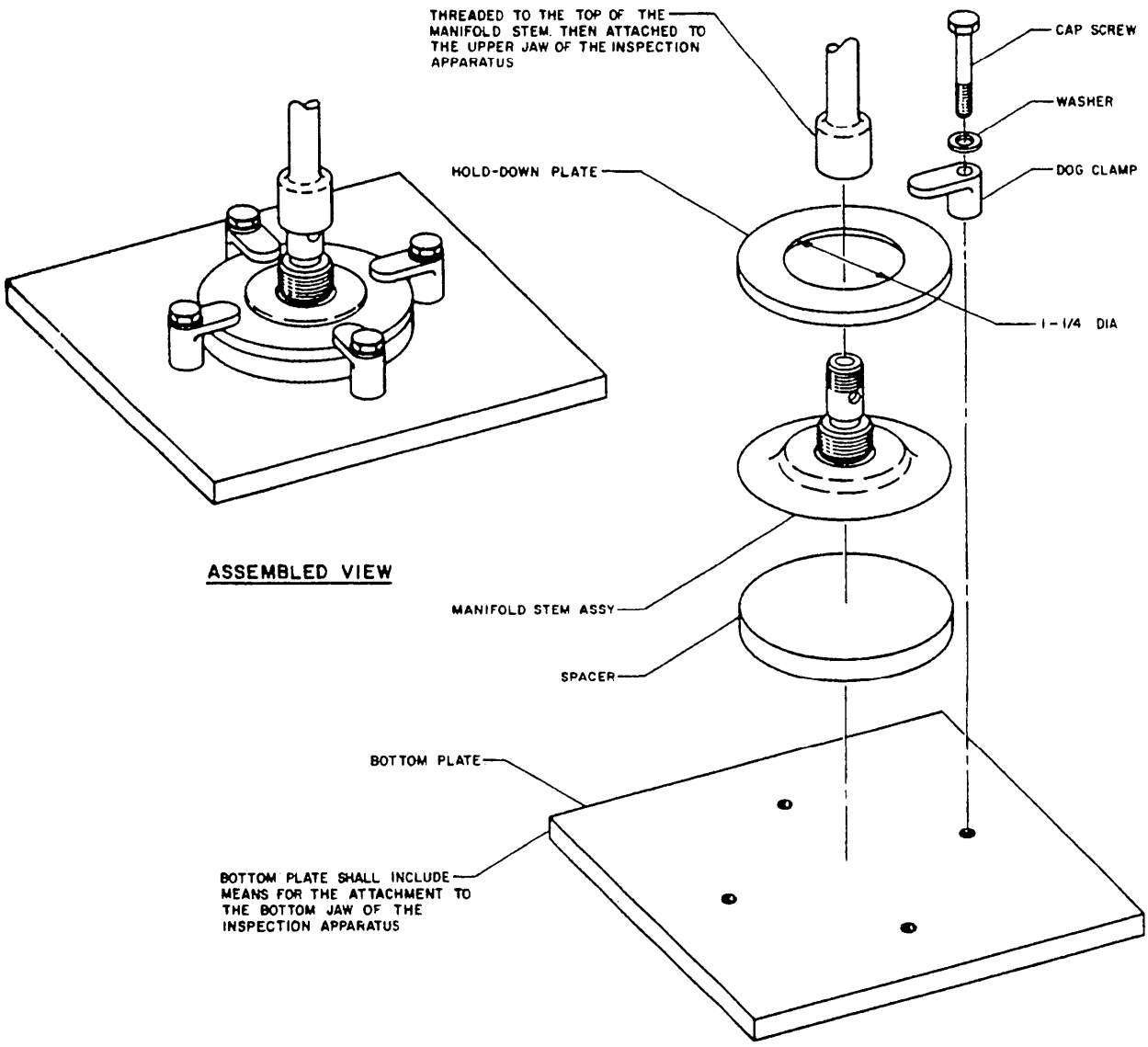



**FIGURE 1.** ADAPTER TO BE USED FOR DETERMINING THE STRENGTH OF THE ATTACHMENT OF THE MOLDED RUBBER BASE TO THE MANIFOLD STEM



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be separated until failure occurs. It is not the intent of this paragraph to restrict the use of the adapter, Figure 1, but rather as suggested apparatus.

4.6.3.2 After aging - The manifold stem assembly shall be exposed in an atmosphere of 170 ± 2 degrees Fahrenheit (77 ± 1 degrees Centigrade) for 168 ± 1/4 hours. The manifold stem assembly shall then be removed and exposed to a temperature of 75 ± 5 degrees Fahrenheit (24 ± 3 degrees Centigrade). After a minimum of 16 hours and not more than 96 hours exposure at 75 ± 5 degrees Fahrenheit, the attachment of the molded robber base to the manifold stem after aging shall be determined as specified in 4.6.3.1. The manifold stem assembly shall be exposed in such a manner that the conditioned air circulates completely around the manifold stem assembly.

4.6.4 Back pressure leakage - This inspection is applicable only when the manifold stem assemblies are purchased with an installed rubber sleeve check valve. The cap nut, gasket, washer, and Type I base nut, as applicable, shall not be installed and without making any adjustment to the check valve, the manifold stem assembly shall be submerged in clean, fresh water having a temperature of 75 ± 5 degrees Fahrenheit (24 ±~3 degrees Centigrade). Oil and liquid water free air pressure of 5.0 ± 0. 1 pounds per square inch gage shall be applied to the outlet side of the manifold stern assembly for 30 seconds. While the pressure is applied, the manifold stem assembly shall be observed for leakage.

4.6.5 Operation - No adjustment to the check valve shall be made, when the manifold stem assembly is purchased with an installed check valve. An inflation assembly conforming to MI-LI-23145, Type I, II, or III shall be installed on the manifold stem assembly. An applicable carbon dioxide filled cylinder or cylinders, as applicable, conforming to MIL-C-601, Type I or II, or MIL--25369, Type II, shall be fully installed in the inflation assembly with hand tightening. The cap nut, gasket, washer, and Type I base nut, as applicable, shall be properly installed on the manifold stem assembly. The carbon dioxide filled cylinders shall be actuated by pulling the lanyard assembly of the inflation assembly in the normal manner, with an abrupt quick motion, and observed as to where the carbon dioxide gas passes into the atmosphere and when applicable, whether the rubber sleeve of the check valve is dislodged. Upon completion the aforementioned, when the manifold stem assembly is purchased with an installed rubber sleeve check valve, the cap nut, gasket, washer, Type I base nut, as applicable, and the inflation assembly shall be removed from the manifold stem assembly and without making any adjustment to the check valve, the manifold stem assembly shall then be subjected to the back pressure leakage, 4.6.4.

4.6.6 Salt spray - The cap nut, gasket, washer, Type I base nut, as applicable, and the check valve, when the valve is a component of the assembly, shall be properly installed on the manifold stem assembly. The outlet side of the stem shall be plugged to prevent the entry of the salt spray during this inspection.



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The manifold stem assembly, prepared as aforementioned, or the rubber sleeve check valve, when purchased individually, as applicable, shall be exposed to 100 hours of salt spray in accordance with FED- STD- 151, Method 811. The rubber sleeve check valve, when purchased individually, or the manifold stem assembly, as applicable, shall be exposed in such a manner that the conditioned air circulates completely around the valve or the manifold stem assembly, as applicable, Upon completion of the salt spray, the check valve or the manifold stem assembly, as applicable, shall be examined for conformance to 3.4.4.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging - Packaging shall be Level A or C, as specified (see 6. 2).

5. 1.1 Level A -

5. 1.1.1 Manifold stem assemblies - The cap nut, gasket, Type I base nut, as applicable, and the rubber sleeve check valve, when the valve is a component of the assembly, shall be properly installed on the manifold stem assembly. Each manifold stem assembly shall be cleaned in accordance with MIL- P-116, Method C-1 and preserved and packaged in accordance with MIL-P- 116, Method 1C-3. No contact preservative is required. A paper label shall be inserted (see 5.3. 1) and immediately after cleaning, each manifold stem assembly shall be heat sealed within a clear polyethylene snug fitting bag of O. 00125 ± 20 percent inch thick material. The heat seal seams shall be straight, continuous, and parallel to each other and the edges of the polyethylene bag, The closing heat seal seam shall be as close as possible to the edge of the bag. Prior to or during the heat sealing operation, excess air shall be exhausted to the extent necessary to permit packaging of the manifold stem assembly in the folding or set-up box, as applicable. The bagged manifold stem assembly shall be suitably cushioned with cellulosic material conforming to PPP- C- 843, Type II, Class A or B and packaged within a snug fitting O. 045 inch thick kraft lined chipboard folding box conforming to PPP- B-566, Variety 1, Style III, Type B, Class a or a snug fitting O. 048 inch thick non-bending set-up box conforming to PPP-B-676, Type I, Variety 1, Class A, Style 4. Box closure shall be effected with 2 inch minimum wide gummed paper tape conforming to PPP-T-45, Type III, Grade A, B, or C. The manifold stem assembly shall not be deformed in the packaged condition nor shall the sealed bag be damaged. Twenty-four unit packages of one type shall be packaged within a snug fitting fiberboard box conforming to PPP-B-636, Type CF, Domestic Class, Variety SW, Grade 275. Closure and sealing of the container shall be in accordance with the appendix to PPP-B-636.

5. 1.1.2 Rubber sleeve check valves (Applicable only when purchased individually) - Each rubber sleeve check valve shall be cleaned in accordance with MIL-P-116, Method C-1 and preserved and packaged in accordance with MILP-116, Method 1 C- 3. No contact preservative is required. A paper label shall be



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inserted (see 5.3. I) and immediately after cleaning, each valve shall be heat sealed within a clear polyethylene snug fitting bag of O. 00125 +20 percent inch thick material. The heat seal seams shall be straight, continuous, and parallel to each other and the edges of the polyethylene bag. The closing heat seal of the bag shall be as close as possible to the edge of the bag. Prior to and during the heat sealing operation, excess air shall be exhausted to the extent necessary to permit packaging of the specified quantity in the fiberboard container. The valve shall not be deformed in the packaged condition nor shall the sealed bag be damaged. One hundred bagged valves shall be packaged with a snug fitting fiberboard box conforming to PPP-B-636, Type CF, Domestic Class, Variety SW, Grade 275. Closure and sealing of the container shall be in accordance with the appendix to PPP-B-636.

5. 1.2 Level C - The manifold stem assemblies or rubber sleeve check valves, as applicable, shall be packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

5.2 Packing- Packing shall be Level A, B, or C, as specified (see 6. 2). Shipping containers, insofar as possible, shall be uniform in size and shape and of minimum cube and tare weight. Each shipping container shall contain manifold stem assemblies of only one type.

5,2.1 Level A - Ninety-six manifold stem assemblies or 400 rubber sleeve check valves (four boxes), as applicable, packaged as specified in 5. 1.1.1 or 5.1.1.2, as applicable, shall be packed as specified in 5.2.2, except that the container shall be Weather Resistant Class, Variety SW, Grade V3C or V3S. Each container shall be constructed, closed, and reinforced with flat steel strapping or tape banding in accordance with the appendix to PPP-B-636.

5.2.2 Level B - Ninety-six manifold stem assemblies or 400 rubber sleeve check valves (four boxes), as applicable, packaged as specified in 5. 1.1.1 or 5.1.1.2, as applicable, shall be packed within a snug fitting fiberboard container conforming to PPP-B-636, Type CF or SF, Domestic Class, Variety SW, Grade 275. Each container shall be constructed and closed in accordance with the appendix to PPP-B-636.

5.2.3 Level C - The packaged manifold stem assemblies or rubber sleeve check valves, as applicable, which require packing for acceptance by the carrier, shall be packed within exterior type shipping containers in a manner that shall insure safe transportation at the lowest rate to the point of delivery. The shipment shall conform to the minimum requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 Marking - In addition to any special marking required by the contract or order, the interior and exterior containers shall be marked in accordante with MIL-STD-129 and shall include the cure date of the elastomeric component



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(month and year) and the type of the manifold stem assembly, as applicable.

5.3.1 Polyethylene bag - Prior to heat sealing, a white paper label with legible and durable black letters and numerals shall be placed inside each polyethylene bag (see 5. 1.1.1 or 5.1.1.2, as applicable). The label shall be placed so that it may be read through the bag and shall contain the same markings as required in 5, 3. The label shall also contain the specification number,

60 NOTES

6.1 Intended use - The manifold stem assemblies with or without an installed check valve covered by this specification are intended for use as a component of the mechanical inflation system for inflatable life preservers.

6.2 Ordering data - Procurement documents shall specify the following:

(a) Title, number, and date of this specification.

(b) Type and quantity desired (see 1. 2),

(c) Whether the manifold stem assemblies are to be furnished with or without an installed rubber sleeve check valve (see 3.3. 2).

(d) Whether the first article inspection is waived (see 4.3. 1),

(e) Name and address of the first article inspection , laboratory (see 4.3. 1).

(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(g) Whether any additional markings are required (see 5. 3),

(h) Certificate of compliance for the age of the elastomeric components (see 3. 2).

6.3 Data - For the information of Contractors and Contracting Officers, any of the data specified in applicable documents listed in Section 2 of this specification, or referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NavWeps Form 4200/15 (Drawings, Lists, and Specifications Required) shall be attached where applicable.



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Custodians:
Army - ME
Navy - AS
Air Force - 11

Preparing activity:
Navy - AS
(Project No. 4220-0079)

Review activities:
Army - ME
Navy - AS and SH
Air Force - 11 and 84

User activities:
Navy - MC

NOTICE - Review/user information is current as of date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof).*

SPECIFICATION

MIL-M-81385A MANIFOLD STEM ASSEMBLIES, LIFE PRESERVER

| ORGANIZATION *(Of submitter)* | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

DD FORM 1 NOV 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279



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FOLD

DEPARTMENT OF THE NAVY
Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Weapons Engineering Standardization Office (Code X)
Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

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GPO 303-223/Z 711

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